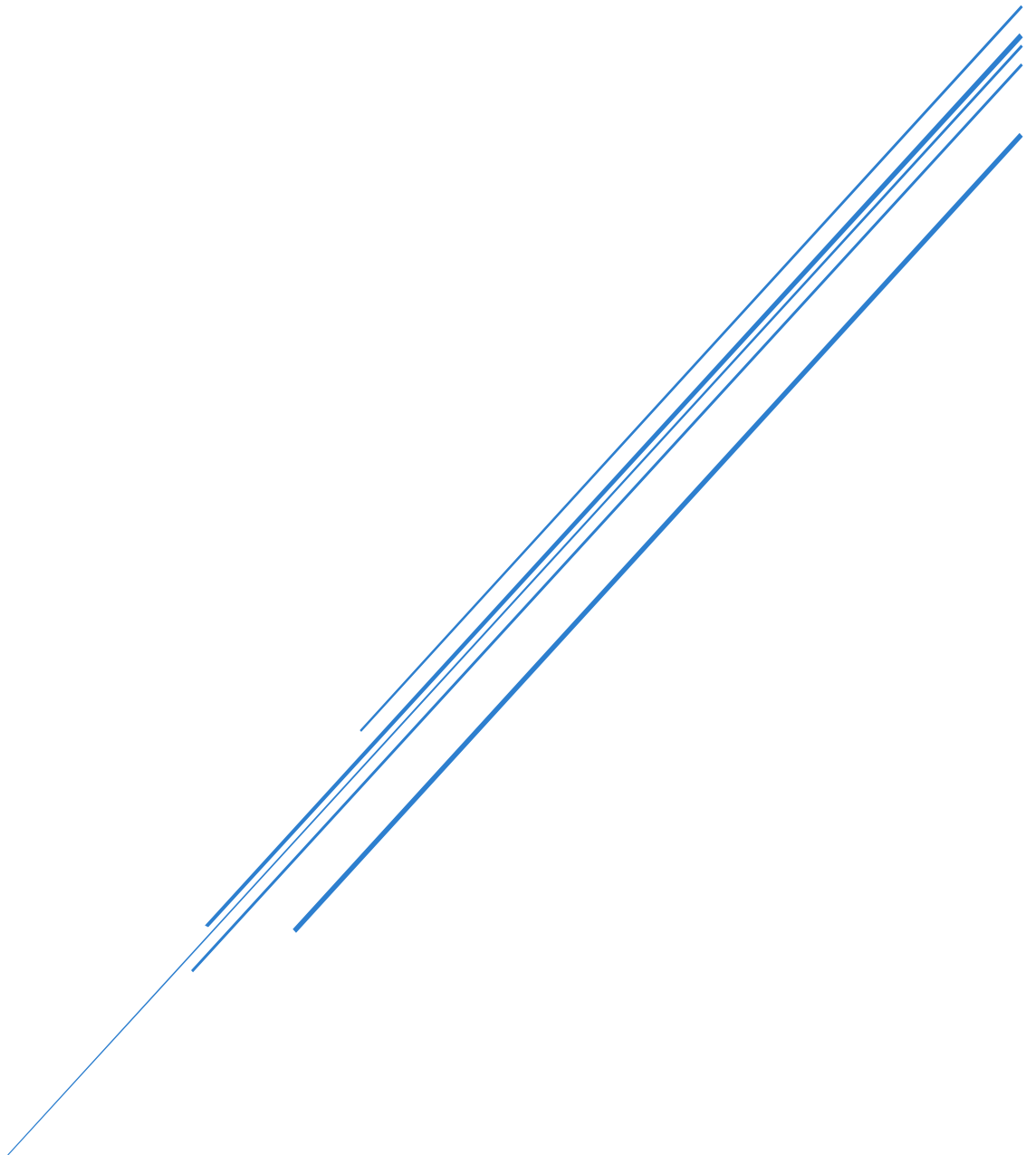


PRINCIPLES OF SECURITY: ASSIGNMENT 1

Full Name: Kandyce Smit

Student Number: ST10467645



Lecturer: Toritseju Oni
Module Code: PRSE6212

Table of Contents

Question 1	2
• Third-Party System Integration and the "Weakest Link" Concept:	2
• Risks from Third Parties Handling Sensitive Data and Systems:	2
• Lack of Expertise, Workarounds, and Incompatible Systems:	3
Question 2	5
2.1)	5
• Brute Force Attack:	5
• Dictionary Attack:	5
• Password Spraying:	5
• Pass the Hash Attack:	5
• Hybrid Attack:	5
2.2)	5
Question 3	7
Question 4	10
References	13

Question 1

The third-party vendor model is growing in popularity among South African companies. Companies will be able to save money and gain specialised knowledge by outsourcing many tasks to third-party vendors, contractors, and service providers. However, as the relationship grows so does the risk of cyber-attacks. The reason for this increased risk is that all third-party vendors and/or contractors will require some level of access to the company's computer network. This increases the risk of a cyber-attack, and the threat is increasing rapidly in South Africa (Corbado , 2026).

- **Third-Party System Integration and the "Weakest Link" Concept:**

Cybersecurity experts believe the weakest link to a business's entire network will be where the business shares a digital connection with a third-party business. This is known as the "weakest link" concept (NIST, 2022). This means that a business's total security is no stronger than the least secure third-party connection they have created, making third party integration a key point of failure in enterprise security. When a company connects to a third party, the third party's system becomes a part of the first business's network. The same applies when a third-party is allowed to connect to a business's internal system. If the third-party does not have good security practices in place (for example, due to a limited budget, few trained IT staff, or poor policies), then all of the third-party's weak security practices are now also a weak security practice for the original business (Syteca, 2025).

In many cases, third-party companies do not have sufficient security measures to protect them from being compromised. Many times, third-party companies have poorly managed systems (i.e., no timely patches), they use older versions of software, and they may not have adequate access control. Once a third-party company's systems are connected to the main business' network, any hacker who compromises the third-party company can gain a potential entry point into the larger organisation (NIST, 2022).

System integration occurs when two or more separate computer systems are linked together so they can both send and receive information. System integration provides a method for connecting multiple systems into a single network; however, this can also provide hackers a way to gain access to the systems once the systems are integrated. An application programming interface (API) is one of the tools that are often used to integrate systems. APIs are simply a way for different software systems to communicate with each other. However, just like any system that communicates over the internet, APIs need to be protected using authentication controls and encryption to prevent unauthorised users from gaining access to the API and using it as a portal to access the rest of the network (Syteca, 2025).

Transnet's 2021 Ransomware Attack Is an Example of Third-Party Network Risk Materialising in South Africa. Transnet is responsible for managing South Africa's ports and freight rail. In 2021, Transnet was attacked by ransomware causing disruptions at every major port in South Africa. Security professionals found that network entry points (those related to third party connected systems) were a significant part of the risk environment that allowed the attack to occur (IT News Africa, 2021). Although the actual entry point used was never disclosed, incidents such as the Transnet ransomware attack illustrate how interconnected systems and third-party access points create opportunities for attackers to expand the organisations attack surface.

Businesses should conduct vendor risk assessments prior to giving a third-party network access. Businesses should also ensure that third parties are given the minimum level of access required (least privilege) and continue to monitor all third-party activity on their networks (NIST, 2022; Syteca, 2025). This illustrates that third-party integrations do not just add another connection to the organisation, they also transfer third party vulnerabilities into the organisation's internal environment, making it one of the largest security threats faced by businesses.

While integrating with other parties can be an efficient means of increasing overall efficiency, it will create a dependency upon a number of external security standards which the organization has no complete control over; this makes mitigating risks as difficult or perhaps impossible, despite having very good internal security practices.

- **Risks from Third Parties Handling Sensitive Data and Systems:**

Beyond enterprise-wide network integration third parties are often given access to sensitive information and critical systems that significantly expose an organisation's risk.

Outsourced Code Development

When a business uses an outside company or individual to develop their software or application, it introduces the risk that there will be vulnerabilities present in the code produced by that developer whether through negligence or deliberate

malicious intent. Outsourced developers may introduce bugs, insecure coding practices, hidden backdoors which allow unauthorised access after deployment (SonarSource, 2024). This is of particular concern when outsourcing takes place in cost-driven environments, such as those common among South African businesses looking to reduce operational costs where proper security vetting and code reviews may not be sufficient or entirely absent (Enlab Software, 2025).

Third-Party Data Storage

Many organisations rely on third party cloud storage providers to manage their data. While this is cost effective and scalable, it means that if a security breach occurs on the provider's side, the organisation's sensitive information could be directly exposed. In these cases, the organisation has limited control over how the provider secures its infrastructure yet still bears full responsibility for any compromised data (Corbado , 2026). South Africa has seen a growing number of data breaches linked to third party storage arrangements, enhancing the need for stricter vendor controls (Corbado , 2026).

Marketing Agencies and Customer Data

Third parties like marketing agencies have access to customers database to run their marketing campaigns. When a third-party experiences a data breach and sensitive customer data is released, the organisation has no direct control over how the data was accessed and has no knowledge of what actions were taken with the data. In terms of regulatory action, POPIA makes it clear that the organisation that collects the data is responsible for the protection of that data (Qualysec, 2026). Therefore, regardless of where the error occurred and whether the third party was at fault, the original organisation could suffer from the same level of regulatory sanctions, civil liability and reputational harm as if the breach had occurred within their own walls. With the addition of POPIA compliance being a requirement for South African organisations, the process of evaluating the security standards of any third party who will handle customer data becomes not only mandatory, but also wise.

The SolarWinds Attack – An example of how large-scale third-party risks can occur

An example of how third parties can create a significant risk for companies is the recent SolarWinds attack which occurred in 2020. An attacker created a piece of malware which was inserted into a SolarWinds software update. The software update was delivered to thousands of customers, some of whom were government agencies and other large corporations. The customers installed the software updates assuming it would fix any problems they were facing with their computers (TechTarget, 2023). This resulted in customers unknowingly allowing the attackers access to their systems as they trusted the source of the software.

These examples illustrate that third parties can increase a company's attack surface, while at the same time increase the potential legal, financial and reputational consequences of a successful cyberattack. Therefore, South African businesses must carefully consider whether to use a third party or not, especially if the third party is going to handle sensitive customer data.

Therefore, even though organisations are using internal security best practices, if one of their third party fails then there could potentially be significant legal and financial consequences for the organisation due to POPIA regulations.

- **Lack of Expertise, Workarounds, and Incompatible Systems:**

The biggest and probably least acknowledged risk for businesses comes from integrating systems that either have compatibility issues with each other, or from staff responsible for the integration being unfamiliar with cybersecurity best practices.

When two systems do not communicate well, IT staff typically develop workarounds: informal, temporary technical solutions to enable communication between systems. The most common workarounds used include deactivating firewalls to allow data to move between systems, the transfer of data without encrypting it, or providing an employee with excessive access to prevent authentication issues (ScienceDirect, 2019). Workarounds typically bypass security controls that have been put into place to protect the network. Since the workarounds are typically non-documented and not formalised, they will be forgotten and provide an attacker with a long-term opportunity to exploit a security gap.

Another contributing factor to the increased risk is the lack of cybersecurity expertise. Due to financial constraints many South African Small Medium Enterprises (SMEs), are unable to hire full-time cybersecurity specialists. Therefore, most SMEs assign the responsibility of performing integration tasks to general IT personnel who may not completely understand the potential security ramifications of their configuration choices (Tech In Africa, 2025). This can lead to poorly configured systems such as open ports on networks, poor encryption settings, or incorrectly implemented Application Programming Interfaces (APIs), all of which can provide an attacker with opportunities to exploit known vulnerabilities.

Legacy systems are also an important consideration. Integrated systems that consist of end-of-life software are extremely vulnerable to attack since they are publicly available and there are active exploits for these systems (NIST, 2022).

Additionally, the use of shadow IT creates additional risks. Shadow IT refers to unauthorised applications or services that employees utilise, typically as a work around when official systems are too complicated or incompatible. Since the organisation has no visibility of the usage of the application or service, they have no ability to monitor the activity of the application or service for security threats (IBM, n.d.).

These combined factors contribute to predictable and identifiable weaknesses in the organisation's systems, enabling the attacker to get around the security measures, elevate their privileges and obtain unauthorised access to the network.

To mitigate these risks, South African businesses must conduct thorough compatibility evaluations prior to implementing third party solutions, have all integration tasks reviewed by qualified cybersecurity experts, and establish and enforce a set of strict policies that prohibit employees from using Shadow IT or other undocumented work-arounds (NIST, 2022).

Thus, while workarounds may help resolve short term operational problems, they greatly decrease long term security and introduce undetectable and unmanageable vulnerabilities.

To conclude, third party partnerships present South African businesses with real and expanding cyber threats. Using the weakest link theory, third party vulnerabilities created by the need for system integration, third party exposure of sensitive information and third party poorly implemented technology create opportunities for attackers to use the third party as an entry point for devastating attacks. The way to protect against these types of threats is to perform appropriate vendor risk assessments, apply access controls, comply with POPIA and perform technically correct implementations of the third-party solution.

Question 2

2.1)

- **Brute Force Attack:**

The Brute Force method is an example of a Trial-And-Error type of attack that hackers employ when they attempt to guess every possible combination of passwords for a particular user ID, until they find the right one (Kaspersky, n.d.; IBM, n.d.).

In the case of South Africa's banking environment, a hacker could potentially utilise automation tools to target the user's online banking account. As many consumers tend to have weak or very short passwords/PINs, it does not take long to break them down and then gain unauthorised entry into their bank account and begin committing financial fraud (FraudNet, n.d.).

- **Dictionary Attack:**

Dictionary attacks are another form of trial-and-error attacks in which a hacker has a pre-existing list of commonly used words and/or passwords to use to gain access to a user's account. This form of attack is different from a Brute Force attack in that it targets the types of passwords that a consumer is most likely to select. (Huntress, 2025).

In South Africa's banking industry, a hacker can use a dictionary attack to target a user who selects predictable passwords, such as a name, common phrase etc. The fact that so many consumers select reusable passwords creates even greater potential for account compromises (FraudNet, n.d.).

- **Password Spraying:**

Password Spraying is a technique employed by hackers where a single commonly used password is attempted across many different accounts to limit the number of logins attempts per account thus avoiding account lockout (CrowdStrike, 2022).

In South Africa's banking sector, a hacker may attempt to access several accounts utilising common passwords. Because the same password(s) are being attempted across numerous users, this creates a larger challenge to identify the hacker, while still enabling unauthorised access to some accounts (SilverFort, n.d.).

- **Pass the Hash Attack:**

Pass the Hash Attacks occur when attackers steal a hashed (encrypted) password and then utilise that hashed password to authenticate to a system or resource. The attackers do not have knowledge of the original password. (TechTarget, 2025).

The ability for attackers in a South African banking environment to utilise stolen hashed passwords as an identity to impersonate legitimate users or employees enables attackers to gain access to systems and financial data. This access is gained without having to "crack" the original password (TechTarget, 2025).

- **Hybrid Attack:**

Hybrid attacks involve modifying common passwords by adding numbers, symbols, or variations. (NETSCOUT, 2024).

Attackers could perform hybrid attacks on the South African banking industry effectively due to the fact many users use slight modifications of common passwords. (e.g., add numbers or symbols). By utilising this behaviour, attackers improve their odds of successfully gaining access to user accounts. (Cyberly, n.d.).

2.2)

The importance of Artificial Intelligence (AI) in Cyber Security continues to increase as well as its use in Banking where account protection of customers is an essential task. AI is used in the banking industry in many important ways:

- **AI Strengthening Password Protection**

Banks have greatly increased the security against password-based attacks with the help of AI. AI Systems can determine the normal behaviour of each customer and will therefore immediately report anything unusual as suspicious (Portnox, 2023). Real-Time anomaly detection is much better than traditional rule-based systems as AI continuously learns and adapts to new dangers and does not only detect known threat patterns (Infosecurity Magazine, 2025).

- **How AI Helps Attackers Crack Passwords**

Cyber Criminals are also provided with highly effective tools by AI. Machine Learning Algorithms can analyse millions of leaked passwords and find patterns of capitalising the first letter of the first word or adding numbers at the end of the password that humans frequently do and prioritise them while trying to crack an account. With AI-powered tools many common passwords can be cracked in just seconds. Therefore, customers of South African banking who make use of short or predictable passwords are at great risk. (Messente, 2025).

- **Alternative Authentication Methods and Whether They Could Replace Passwords**

Biometric authentication, as well as behavioural authentication both provide improved options over passwords due to AI. Biometric authentication uses AI to authenticate an individual based upon their biometric data, which is very difficult to guess, or steal compared to a password (Identity.org, 2025). Behavioural authentication takes this a step further by utilising AI to assess how you type, hold your phone, and your normal spending habits to continue verifying your identity as a "background" check (Portnox, 2023). Although biometric and behavioural authentication are superior to traditional password authentication, none of them will completely replace passwords. Deep fake technology has already been developed to deceive facial recognition and voice verification systems so a Multi-Factor Authentication approach will still be the best method for banking to ensure the highest level of security (Identity.org, 2025).

In addition to all the progress made, entirely eliminating passwords as an access method will likely not be possible in the near future because of costs related to password less technology, difficulties with users adopting new methods, and potential risks associated with biometrics, reinforcing the need for multiple forms of identification.

- **South African Banks Using AI-Driven Security**

South Africa has been at the forefront of utilising AI driven security solutions. Standard Bank utilises AI systems to identify, monitor and stop fraudulent payments in real-time. The AI analyses payment patterns for any anomalies (Standard Bank, 2024). In addition to preventing payment fraud, South African banks have been identified as leaders in the development of AI-based anti-money laundering solutions. Machine learning is utilised to identify abnormal financial transactions that are indicative of criminal behaviour (Fintech Global, 2025). South African banks are developing AI driven security solutions that go beyond regulatory requirements and proactively utilise AI to prevent financial crime from occurring (SymphonyAI, 2025).

Overall, AI is enhancing authentication within the banking industry, however, the complexity of cyber-attacks is also being enhanced by AI. As a result, banks must continually update their security strategies to keep up with evolving threats.

Question 3

1. Executive Summary

This report assesses various forms of firewalls to identify the appropriate solution for The African Taste Co., a growing South African SMME engaged in the e-commerce sector. Firewalls are network security systems designed to monitor and control incoming and outgoing network traffic according to established security policies (Fortinet, n.d.). With The African Taste Co.'s expansion into online commerce and the increasing volume of sensitive customer information being processed through these operations, the potential for cyber-attacks to occur has dramatically increased.

This report will compare the primary characteristics of four types of firewalls: stateful versus stateless, open-source versus proprietary, hardware versus software, and host-based versus virtual firewalls. A Fortinet FortiGate 60F is recommended as the most suitable option to meet The African Taste Co.'s needs for scalability, protection and affordability due to the nature of their business.

2. Background

The African Taste Co. is a South African Small to Medium Enterprise (SMME) employing around fifty employees and generating an annual revenue of R10 Million. The business specialises in exporting foods and recently entered the e-commerce space where they process online transactions and store customer information. These new activities have greatly expanded the scope of cyber-security risks facing the business, particularly data breaches, unauthorised access and online fraudulent activity.

As The African Taste Co. continues to grow so too does the complexity of their network creating greater interest from cyber-criminals who view them as targets. Therefore, a firewall is required at this stage of development as it serves as the first layer of defence against malicious traffic attempting to enter or leave their networks (FCA Africa, 2025). In addition to acting as an initial layer of defence for an organisations' computer networks, today's firewalls often contain additional capabilities including; Virtual Private Network (VPN), Intrusion Prevention System (IPS) and Content Filtering (Enterprise Networking Planet, 2023).

3. Firewall Categories Considered

• Stateful vs. Stateless Firewalls

Stateful firewalls are considered more advanced than stateless firewalls because they monitor all current connections. Therefore, they understand the context of network traffic. Stateful firewalls allow for better detection of unusual or suspicious behaviour and provide a greater sense of security (Fortinet, n.d.). However, stateful firewalls are more expensive and require more processing power.

Stateless firewalls, unlike stateful firewalls, do not monitor the connections that occur. Instead, they look at each individual data packet and evaluate it according to a set of predetermined rules. While this method provides speedier results and lower costs compared to stateful firewalls, they lack the ability to identify patterns or abnormal behaviours, which means their security features are limited (ConnectWise, 2025).

To meet its needs as an e-commerce company handling customers' information online, The African Taste Co. would benefit from using a stateful firewall due to the need for higher levels of security.

• Open Source vs. Proprietary Firewalls

Firewalls offered as open source are available without charge. Users also have the option to modify the code. Although these types of firewalls save money and provide flexibility, proper configuration and maintenance requires significant expertise from IT professionals (EM360Tech, 2024).

Proprietary firewalls are created and supported by companies. These often feature technical support, frequent updates and user-friendly interfaces. A disadvantage of proprietary firewalls is that there is typically an additional licensing fee involved (Undercode Testing, 2025).

The African Taste Co. could take advantage of proprietary firewalls since it is an SMME and may not have access to specialised IT personnel. As such, proprietary firewalls will be much simpler to manage and include the availability of professional assistance.

• Hardware vs. Software Firewalls

Hardware firewalls are typically implemented using physical devices that are placed between an organisation's network and the Internet. These types of firewalls offer full protection of all data on the network, making them more suited for larger organisations with centralised networks (Palo Alto Networks, n.d.).

Firewalls that utilise software are usually installed individually on each system within an organisation. Although these solutions are much more cost-effective and simpler to install than their hardware counterparts, they do not offer the same type of comprehensive protection for an entire organisation's network (Fortinet, n.d.).

Therefore, given that The African Taste Co. has approximately fifty people who work out of a common network environment, it would make sense to use a hardware firewall to protect the entire network from a single location.

- **Host-Based vs. Virtual Firewalls**

Host-based firewalls are pieces of software that run on individual computers. They protect at the device level. Host-based firewalls could be good to add to the overall security posture of an organisation, but managing multiple host-based firewalls can be very time consuming and expensive (NetSecCloud, 2024).

Virtual firewalls are pieces of software that run in virtualised/cloud environments. They provide flexibility and scalability and do not require the purchase of any hardware. Because of this, virtual firewalls are especially useful for organisations that depend heavily on cloud computing (TechTarget, 2024).

In addition to using a hardware firewall, host-based firewalls could be implemented on certain devices as another layer of security by The African Taste Co. If, however, the organisation decides to expand its operations and move to a cloud-based model in the future, virtual firewalls may be more beneficial.

Although each type of firewalls has their own strengths, no one type can provide full protection by itself; therefore, there are several considerations when choosing the right firewall for your organisation including cost, how secure you want your organisation to be, how easily you can manage the firewall system and what environment your organisation operates in.

4. Recommendation

Considering the analysis of firewall categories mentioned previously, I recommend that The African Taste Co. should install the Fortinet FortiGate 60F NGFW as its firewall. This product is designed to meet the needs of small and midsize business, like The African Taste Co., and is commonly found in SMME environments (Fortinet, n.d.; Balanced+, 2025).

Why the FortiGate 60F Suits the African Taste Co:

The African Taste Co. employs approximately fifty workers who work out of one location so when it comes to the network configuration, a hardware-based firewall provides better security for a business using a shared network. It will protect the whole network via one hardware device instead of having to install and manage the software on every machine (Palo Alto Networks, n.d.); whereas a software based solution will not provide the organisation wide security required by a company of its size.

The African Taste Co. conducts their business through an e-commerce platform and collects sensitive information regarding their customers; this creates a higher amount of cyber risk on a day-to-day basis. This makes a stateful firewall necessary because it continuously monitors and records all current active connection states within the network allowing for the identification and detection of new or abnormal activity patterns in real-time (Fortinet, n.d.); as opposed to a stateless firewall that only examines each packet individually looking at them against pre-defined rules and does not possess the ability to identify complex evolving threat types.

Since The African Taste Co. is an SMME they probably do not have a full-time staffed Cybersecurity Team, utilising a proprietary firewall like the FortiGate 60F will be much simpler to use and maintain versus an open-source option. An open-source firewall such as pfSense will need experienced IT Professionals to configure and administer properly, if they don't have those skills then the firewall could be configured improperly exposing the network (Undercode Testing, 2025). The FortiGate 60F is a proprietary product providing automatic and frequent security updates along with an easy-to-use interface, eliminating the requirement for technical expertise to run the Firewall (Fortinet, n.d.).

Key Features of the FortiGate 60F:

The FortiGate 60F delivers a full range of security features that meet directly with the needs of The African Taste Co. (Fortinet, n.d.; Dell, n.d.)

- Stateful inspection - tracks all active connections and identifies threats based on context and behaviour.
- Intrusion Prevention System (IPS) - actively prevents attacks by monitoring and blocking both known and unknown attack types.
- Content filtering - Blocks employee's access to malicious or inappropriate websites.
- VPN support - Allows remote workers to connect securely with the company network while working off-site, especially now that the business has grown so much.

- AI-powered FortiGuard security services - Continuously receives updates from its Threat Intelligence system to identify and respond to new threats in real-time.
- SD-WAN capability - Optimises network speeds for companies like The African Taste Co. who operate primarily online through their e-commerce website.

Cost Consideration:

Based on the approximate price of R8000-R15000, the FortiGate 60F is affordable and justifiable for a business making an annual income of R10 million (Fortinet, n.d.). The financial risk of a single data breach, loss of customer confidence, regulatory fines due to non-compliance with POPIA regulations, and lost revenue due to business downtime will significantly outweigh the costs associated with purchasing the FortiGate 60F firewall (FCA Africa, 2025). Therefore, the purchase of this firewall is not an optional expense for The African Taste Co. as it is essential for ensuring their ongoing success and development of e-commerce capabilities.

Why Not the Alternatives:

To explain why the alternative options listed above were not considered, these are some of the reasonings:

- Open-source firewalls were eliminated due to the requirement for specific technical knowledge and experience by The African Taste Co. who currently do not have these skills.
- Software firewalls were rejected as software firewalls only protect each device individually and therefore cannot provide centralised protection for the complete business network.
- Stateless firewalls were eliminated as they are not intelligent enough nor able to make contextual decisions regarding protecting their e-commerce platform from contemporary cybersecurity threats.
- (ConnectWise, 2025; Fortinet, n.d.).

As a result, the FortiGate 60F offers the best combination of security, affordability and usability to meet The African Taste Co.'s current and future requirements. The choice of which firewall system to use was primarily made using technical capabilities, however, since this organisation does not have any internal expertise related to cybersecurity, the ability to effectively manage a particular system became a critical consideration so that the organisation could continue to function securely over time.

5. Conclusion

This report covered the characteristics of different Firewall types, namely: Stateful vs. Stateless; Open Source vs. Proprietary; Hardware vs. Software; Host Based vs. Virtual Firewalls. The Review assessed each type on their pros and cons, with respect to The African Taste Co. and their growing E-Commerce operation. The report revealed that a proprietary Hardware Firewall using a stateful approach would offer the best possible security and thus the preferred option for businesses dealing with highly confidential customer data (Fortinet, n.d.; ConnectWise, 2025)

Based upon the previous review, the Fortinet FortiGate 60F was identified as the most suitable solution based on its network wide protection capabilities, advanced security long term management, and cost effectiveness for an SMME environment (Fortinet, n.d.). Implementing a FortiGate will provide The African Taste Co. with necessary protections to safeguard the company's network, protect customers trust and ultimately support the long-term growth and success of the company (FCA Africa, 2025).

Therefore, determining which firewall is best suited for your business is not just a technical decision, but rather a strategic business decision that will affect the organisation's overall ability to do business safely and ultimately impact their customers' confidence.

Question 4

1. Overview of TPM and Key Functionalities

Trusted Platform Modules (TPMs) are relatively new to computing. A TPM is a small piece of hardware installed in computers, laptops, tablets etc. The primary function of a TPM is to enhance security through secure storage of sensitive data, for example passwords, encryption keys and digital certificates, essentially working as a "Digital Lock Box" within the device that protects sensitive data from being copied, removed or accessed without authorisation (Microsoft, 2025; IBM, 2026).

Since a TPM stores data at a hardware level, this makes it very difficult for malicious users to circumvent or bypass when compared to traditional software-based security methods.

Some of the main functionalities provided by a TPM include:

- **Cryptographic services:** A TPM can encrypt and decrypt data. Data encryption takes sensitive information, like your credit card number or Social Security number, and makes it readable only to those authorised to do so (PDQ, 2025).
- **Key generation and secure storage:** A TPM creates secure encryption keys and then stores these keys securely inside the TPM. These keys will be used for securing data, authenticating devices, and creating digital signatures (IBM, 2026).
- **System integrity checks (Secure Boot):** The TPM verifies that the system has not been altered during boot up. It verifies that the operating system and firmware are valid and secure prior to allowing full operation (Microsoft, 2025).
- **Device authentication:** A TPM will allow a device to verify itself to a network or system before being granted access, thereby preventing unauthorised devices from accessing resources (TechTarget, 2022).

In general, a TPM supplies a robust hardware-based security base for protecting both the device and its data.

2. Specific Benefits of TPM for South African Home Users

The use of TPM technology is an excellent option for South African home users due to the large amount of device theft and cybercrime in South Africa.

As mentioned earlier many common home devices such as laptops and tablets store sensitive personal information such as bank account numbers, identity documents, email accounts and private file systems. TPM protects this type of information from unauthorised access through the requirement of proper authentication.

Some of the main benefits of using TPM include:

- **Better Data Protection:** TPM encodes sensitive data so that no one, but the intended recipient can decode/read the data (TechTarget, 2022).
- **Protection against unauthorised access:** Although an individual may obtain physical access to a device that contains protected data, a TPM will still ensure that they cannot view the data without first obtaining proper authorisation (Microsoft, 2025).
- **Protection of financial and personal information:** With the growing use of online banking and digital transactions in South Africa, it is even more critical to protect your financial and personal information (Investec, n.d.).
- **Device integrity verification:** TPM verifies that there have been no alterations made to the System, and therefore no malware introduced into the system prior to booting up and thus reduces potential cyber threats (Microsoft, 2025).
- **Support for secure authentication:** TPM can support secure login authentication utilising PIN's, Passwords, and Biometric data, such as fingerprint or facial recognition data, to enhance login security (Microsoft, 2025).

As such, TPM offers protection against these common forms of crime in South Africa, including theft, burglary and cyber-crime which are all major concerns when protecting data. The success of TPM is therefore heavily dependent upon how well it is both activated and utilised by the user; otherwise, the advantages of TPM can only be achieved if a combination of best security practices are being followed.

3. Mitigating Risks from Theft and Unauthorised Access

There are significant opportunities to mitigate many of the risks related to theft and unauthorised access to sensitive information through TPM.

The most direct way TPM provides protection against unauthorised access to sensitive information is through encryption. Once a device has been secured with TPM, TPM will encrypt all sensitive files on the device with encryption keys that are securely stored within the TPM Chip. Therefore, even if a laptop or tablet is stolen, the data contained within the device cannot be accessed without proper authentication credentials (Microsoft, 2025).

Additionally, the TPM prevents unauthorised access through several methods:

- Data remains encrypted: Files stay encrypted so no one will be able to read your data unless they have the correct decryption key to unlock the file. The decryption key is stored on the TPM.
- Hardware-based key protection: The encryption keys are stored within the TPM. It is very difficult to extract, copy or obtain these keys by an attacker. That makes it virtually impossible to circumvent your TPM based protections.
- System integrity verification: On start-up, the TPM verifies if the system has been altered. If any suspicious alterations or malware are identified, the system may deny you access to the device or even fail to completely boot up (IBM, 2026).
- Secure login and authentication: TPM works along with your login method like PINs, Passwords, Biometric scanning etc., so only authorised people can use your device.
- Protection against credential theft: Sensitive credentials are stored safely on hardware; therefore, an attacker cannot easily steal or reuse this information.

In cases where devices are stolen, which is common in South Africa, TPM provides assurance that while a thief may be able to activate your device, he/she will not be able to access your financial/identity information. Thus, reducing the damage caused by theft and protecting your identity and information (TechTarget, 2022). Thus, TPM provides strong protections for data at rest however should be paired with additional security measures so as to provide comprehensive security throughout an active device's lifecycle.

4. Recommendations for Home Users on Implementing TPM

Enabling TPM on South African home-user devices does not require specialised technical knowledge; therefore, TPM is an accessible and practical method of securing that system. To begin using TPM you need to verify if your device contains TPM by entering "tpm.msc" into the Windows search bar. If the TPM Management window appears, then your device contains TPM 2.0 (Microsoft Support, n.d.).

Next, when TPM is disabled, you will need to reboot the device, go into the BIOS by hitting F2 or delete while booting, locate the security option and then select enable TPM (Microsoft Support, n.d.).

After that you can turn-on BitLocker Drive Encryption via the Windows control panel. Once enabled BitLocker utilises the TPM on the chip to fully encrypt the hard drive to ensure that no unauthorised individual may read any information that was written to the drive (Microsoft Learn, n.d.). This is especially useful in South Africa due to widespread device theft, as a BitLocker-enabled stolen laptop will make itself unusable to thieves.

When activating BitLocker, it will generate a recovery key. It is essential to store this key safely at a remote location, such as a Microsoft Account, print a copy and/or store it on a removable USB (Microsoft Learn, n.d.). Loss of the recovery key will permanently deny authorised users access to their data.

Finally, users can further secure their systems by enabling a pre-boot pin. The pre-boot pin will prevent someone from accessing the device until the operating system is loaded. Additional features such as Windows Hello, such as fingerprint login or facial recognition, can be utilised to improve authentication. Users should continue to update both their system and firmware with current versions as well to ensure that any issues related to TPM vulnerabilities are resolved (Microsoft Learn, n.d.).

Overall, TPM provides a simple-to-use and easily implementable form of security protection for home-users who do not possess special technical knowledge.

5. Limitations and Challenges of Using TPM

TPM offers many advantages, but there are some significant disadvantages that South African home users need to be aware of.

The first limitation is that TPM primarily secures data "at rest." Once you unlock your device and start using it, TPM does not encrypt the data. As a result, there is always a possibility for malware and/or other types of cyber-attacks to happen (Microsoft Learn, 2026). Therefore, TPM should always be used in conjunction with an anti-virus program and good password practices.

Another disadvantage is if the user loses the BitLocker Recovery Key. The user will then be unable to recover access to their device and all data saved on that device will be unrecoverable (Microsoft Learn, 2026).

There is also a concern regarding hardware compatibility. A lot of older or second-hand devices that are commonly purchased by South African homeowners do not have TPM 2.0 capabilities. This means the users cannot take advantage of the enhanced security provided by TPM (Microsoft Learn, 2021).

Low digital literacy and lack of technical skills is another issue faced in South Africa. Many people are unaware of TPM and those who are aware, have difficulty activating TPM. As a result of this, TPM is rarely activated even when it is enabled (Afrobarometer, 2025; GCRO, 2026; Springer, 2025).

Other environmental factors can also contribute to TPM-related challenges. When unexpected power-outages due to load-shedding occur during TPM-enabled boot-up processes, the TPM verification process may fail resulting in startup problems (Microsoft Learn, 2026).

It is extremely unlikely that TPM can be successfully attacked by hackers. However, there are some instances where hackers with advanced knowledge of hardware-based hacking can potentially target TPM. Although, these attempts are rare (Security StackExchange, 2024).

While TPM provides excellent protection from cyber threats, TPM should never be relied upon as the sole method of protecting against cyber threats. TPM should be viewed as one component of a larger overall cybersecurity plan that incorporates education about cybersecurity and safe computer practices, secure login/registration methods and use of multiple security programs to protect computers. Hence, although TPM greatly increases the level of device security, it should not be viewed as a complete solution, but as one piece of a larger Cybersecurity strategy.

References

Afrobarometer, 2025. *PP95: Africa's digital divide is closing, but participation in the digital(ised) economy remains highly uneven*. [Online]

Available at: <https://www.afrobarometer.org/publication/pp95-africas-digital-divide-is-closing-but-participation-in-the-digitalised-economy-remains-highly-uneven/>

[Accessed 26 March 2026].

Balanced+, 2025. *Comparing the Fortigate 40F, 60F, and 80F*. [Online]

Available at: <https://balanced.plus/choosing-your-fortigate-smb-firewall-comparison-40f-60f-80f/>

[Accessed 24 March 2026].

ConnectWise, 2025. *Stateful vs. stateless firewall*. [Online]

Available at: <https://www.connectwise.com/blog/stateful-vs-stateless-firewall>

[Accessed 23 March 2026].

Corbado , 2026. *10 Biggest Data Breaches in South Africa [2026]*. [Online]

Available at: <https://www.corbado.com/blog/data-breaches-south-africa>

[Accessed 17 March 2026].

CrowdStrike, 2022. *Password Spraying*. [Online]

Available at: <https://www.crowdstrike.com/en-us/cybersecurity-101/cyberattacks/password-spraying/>

[Accessed 19 March 2026].

Cyberly, n.d. *What Is A Hybrid Attack In Password Cracking?*. [Online]

Available at: <https://www.cyberly.org/en/what-is-a-hybrid-attack-in-password-cracking/>

[Accessed 19 March 2026].

Dell, n.d. *Fortinet FortiGate FG-60F Network Security Firewall Appliance*. [Online]

Available at: <https://www.dell.com/en-us/shop/fortinet-fortigate-fg-60f-network-security-firewall-appliance/apd/aa881028/wifi-and-networking>

[Accessed 24 March 2026].

EM360Tech, 2024. *Top 10 Best Open-Source Firewalls for 2024*. [Online]

Available at: <https://em360tech.com/top-10/open-source-firewalls>

[Accessed 23 March 2026].

Enlab Software, 2025. *The Role of Cybersecurity in Outsourced Software Development Projects*. [Online]

Available at: <https://enlabsoftware.com/offshore-software-development/the-role-of-cybersecurity-in-outsourced-software-development-projects.html>

[Accessed 17 March 2026].

Enterprise Networking Planet, 2023. *8 Best Firewalls for Small & Medium Business (SMB) Networks*. [Online]

Available at: <https://www.enterprisenetworkingplanet.com/guides/best-firewalls-for-small-medium-business/>

[Accessed 23 March 2026].

FCA Africa, 2025. *Firewalls for SMEs: Your First Line of Defense Against Cyber Threats*. [Online]
Available at: <https://www.fcaafrica.com/post/firewalls-for-smes-your-first-line-of-defense-against-cyber-threats>
[Accessed 23 March 2026].

Fintech Global, 2025. *How South African banks lead AML innovation with AI*. [Online]
Available at: <https://fintech.global/2025/07/14/how-south-african-banks-lead-aml-innovation-with-ai/>
[Accessed 20 March 2026].

Fortinet, n.d. *FortiGate FortiWiFi 60F Series Data Sheet*. [Online]
Available at: <https://www.fortinet.com/resources/data-sheets/fortigate-fortiwifi-60f-series>
[Accessed 24 March 2026].

Fortinet, n.d. *Next-Generation Firewall (NGFW)*. [Online]
Available at: <https://www.fortinet.com/products/next-generation-firewall>
[Accessed 24 March 2026].

Fortinet, n.d. *Stateful vs Stateless Firewalls: Differences And Examples*. [Online]
Available at: <https://www.fortinet.com/resources/cyberglossary/stateful-vs-stateless-firewall>
[Accessed 23 March 2026].

Fortinet, n.d. *What Is A Firewall?*. [Online]
Available at: <https://www.fortinet.com/resources/cyberglossary/firewall>
[Accessed 23 March 2026].

Fortinet, n.d. *What Is A Hardware Firewall? Hardware vs Software Firewalls*. [Online]
Available at: <https://www.fortinet.com/resources/cyberglossary/hardware-firewalls-better-than-software>
[Accessed 23 March 2026].

FraudNet, n.d. *Credential Stuffing*. [Online]
Available at: https://www.fraud.net/glossary/credential-stuffing?utm_source=chatgpt.com#credential-stuffing-statistics
[Accessed 19 March 2026].

GCRO, 2026. *Digital divide in Gauteng*. [Online]
Available at: <https://gcro.ac.za/outputs/map-of-the-month/detail/digital-divide-in-gauteng/>
[Accessed 26 March 2026].

Huntress, 2025. *What is a Dictionary Attack?*. [Online]
Available at: <https://www.huntress.com/cybersecurity-101/topic/what-is-dictionary-attack>
[Accessed 19 March 2026].

IBM, 2026. *Trusted platform module (TPM) in IBM Power*. [Online]
Available at: <https://developer.ibm.com/articles/trusted-platform-module-power-systems/>
[Accessed 25 March 2026].

IBM, n.d. *What is a brute force attack?*. [Online]
Available at: <https://www.ibm.com/think/topics/brute-force-attack>
[Accessed 19 March 2026].

IBM, n.d. *What is shadow IT?*. [Online]

Available at: <https://www.ibm.com/think/topics/shadow-it>

[Accessed 08 March 2026].

Identity.org, 2025. *The Intersection of Artificial Intelligence (AI) and Biometrics*. [Online]

Available at: <https://www.identity.org/the-intersection-of-artificial-intelligence-ai-and-biometrics/>

[Accessed 20 March 2026].

Infosecurity Magazine, 2025. *Can AI Help Protect Passwords from Hackers?*. [Online]

Available at: <https://www.infosecurity-magazine.com/blogs/can-ai-help-protect-passwords-from/>

[Accessed 20 March 2026].

Investec, n.d. *Cybersecurity: Trends, threats, and strategy*. [Online]

Available at: https://www.investec.com/en_za/focus/money/cybersecurity-in-south-africa-2024.html

[Accessed 25 March 2026].

IT News Africa, 2021. *Transnet Undergoes Apparent Ransomware Hack*. [Online]

Available at: <https://www.itnewsafrika.com/2021/07/transnet-undergoes-apparent-ransomware-hack/>

[Accessed 17 March 2026].

Kaspersky, n.d. *Brute Force Attack: Definition and Examples*. [Online]

Available at: <https://www.kaspersky.com/resource-center/definitions/brute-force-attack>

[Accessed 19 March 2026].

Messente, 2025. *How Quickly Can AI Crack Your Password?*. [Online]

Available at: <https://messente.com/blog/ai-password-cracking-2025>

[Accessed 20 March 2026].

Microsoft Learn, 2021. *Windows 11 TPM Disadvantages ?*. [Online]

Available at: <https://learn.microsoft.com/en-us/answers/questions/4253248/windows-11-tpm-disadvantages>

[Accessed 26 March 2026].

Microsoft Learn, 2026. *BitLocker and TPM: other known issues*. [Online]

Available at: <https://learn.microsoft.com/en-us/troubleshoot/windows-client/windows-security/bitlocker-and-tpm-other-known-issues>

[Accessed 26 March 2026].

Microsoft Learn, n.d. *TPM recommendations*. [Online]

Available at: <https://learn.microsoft.com/en-us/windows/security/hardware-security/tpm/tpm-recommendations>

[Accessed 26 March 2026].

Microsoft Support, n.d. *Enable TPM 2.0 on your PC*. [Online]

Available at: <https://support.microsoft.com/en-us/windows/enable-tpm-2-0-on-your-pc-1fd5a332-360d-4f46-a1e7-ae6b0c90645c>

[Accessed 26 March 2026].

Microsoft, 2025. *How Windows uses the Trusted Platform Module*. [Online]
Available at: <https://learn.microsoft.com/en-us/windows/security/hardware-security/tpm/how-windows-uses-the-tpm>
[Accessed 25 March 2026].

Microsoft, 2025. *Trusted Platform Module Technology Overview*. [Online]
Available at: <https://learn.microsoft.com/en-us/windows/security/hardware-security/tpm/trusted-platform-module-overview>
[Accessed 25 March 2026].

NETSCOUT, 2024. *What is a Hybrid Attack?*. [Online]
Available at: <https://www.netscout.com/what-is/hybrid-attack>
[Accessed 19 March 2026].

NetSecCloud, 2024. *The Pros and Cons of Host-Based Firewalls*. [Online]
Available at: <https://netseccloud.com/the-pros-and-cons-of-host-based-firewalls>
[Accessed 23 March 2026].

NIST, 2022. *Cybersecurity Supply Chain Risk Management Practices for Systems and Organizations*. [Online]
Available at: <https://csrc.nist.gov/pubs/sp/800/161/r1/final>
[Accessed 17 March 2026].

Palo Alto Networks, n.d. *Hardware Firewalls vs. Software Firewalls: A Comparison*. [Online]
Available at: <https://www.paloaltonetworks.com/cyberpedia/hardware-firewall-vs-software-firewall>
[Accessed 23 March 2026].

PDQ, 2025. *What is a TPM (Trusted Platform Module)?*. [Online]
Available at: <https://www.pdq.com/blog/what-is-a-tpm/>
[Accessed 25 March 2026].

Portnox, 2023. *AI-Powered Identity Authentication Is Here: What You Need To Know*. [Online]
Available at: <https://www.portnox.com/blog/network-security/ai-powered-identity-authentication/>
[Accessed 20 March 2026].

Qualysec, 2026. *South Africa: A Complete Guide for Businesses*. [Online]
Available at: <https://qualysec.com/popia-compliance/>
[Accessed 17 March 2026].

ScienceDirect, 2019. *Understanding Theory of Workarounds in Practice*. [Online]
Available at: <https://www.sciencedirect.com/science/article/pii/S1877050919318241>
[Accessed 18 March 2026].

Security StackExchange, 2024. *Do a TPM's benefits outweigh the risks?*. [Online]
Available at: <https://security.stackexchange.com/questions/187820/do-a-tpms-benefits-outweigh-the-risks>
[Accessed 26 March 2026].

SilverFort, n.d. *Credential Stuffing*. [Online]
Available at: <https://www.silverfort.com/glossary/credential-stuffing/>
[Accessed 19 March 2026].

SonarSource, 2024. *5 Risks of Outsourcing Software Development and How to Avoid Them*. [Online]

Available at: <https://www.sonarsource.com/blog/5-risks-of-outsourcing-software-development-and-how-to-avoid-them/>

[Accessed 17 March 2026].

Specops, 2025. *AI arms race: How AI will be used by cyber-attackers (and defenders)*. [Online]

Available at: <https://specopssoft.com/blog/ai-in-cybersecurity-arms-race-attackers-defenders/>

[Accessed 20 March 2026].

Springer, 2025. *Bridging the digital divide: exploring the challenges and solutions for digital exclusion in rural South Africa*. [Online]

Available at: <https://link.springer.com/article/10.1007/s44282-025-00189-2>

[Accessed 26 March 2026].

Standard Bank, 2024. *The role of AI in detecting, monitoring, and preventing payments fraud*. [Online]

Available at: <https://www.standardbank.co.za/southafrica/news-and-media/newsroom/the-role-of-ai-in-detecting-monitoring-and-preventing-payments-fraud>

[Accessed 20 March 2026].

SymphonyAI, 2025. *Beyond compliance: South African banks pioneer financial crime prevention*. [Online]

Available at: <https://www.symphonyai.com/resources/blog/financial-services/compliance-south-african-banks-financial-crime-prevention/>

[Accessed 20 March 2026].

Syteca, 2025. *Third-Party Security Risks: How to Mitigate Potential Cybersecurity Threats*. [Online]

Available at: <https://www.syteca.com/en/blog/third-party-providers>

[Accessed 17 March 2026].

Tech In Africa, 2025. *Cybersecurity Challenges for African SMEs*. [Online]

Available at: <https://www.techinafrica.com/cybersecurity-challenges-for-african-smes/>

[Accessed 18 March 2026].

TechTarget, 2022. *Trusted Platform Module (TPM)*. [Online]

Available at: <https://www.techtarget.com/whatis/definition/trusted-platform-module-TPM>

[Accessed 25 March 2026].

TechTarget, 2023. *SolarWinds hack explained: Everything you need to know*. [Online]

Available at: <https://www.techtarget.com/whatis/feature/SolarWinds-hack-explained-Everything-you-need-to-know>

[Accessed 17 March 2026].

TechTarget, 2024. *virtual firewall*. [Online]

Available at: <https://www.techtarget.com/searchsecurity/definition/virtual-firewall>

[Accessed 23 March 2026].

TechTarget, 2025. *What is a pass-the-hash attack?*. [Online]

Available at: <https://www.techtarget.com/searchsecurity/definition/pass-the-hash-attack>

[Accessed 19 March 2026].

Undercode Testing, 2025. *Enterprise Firewalls vs Open Source: A Comprehensive Comparison*.
[Online]
Available at: <https://undercodetesting.com/enterprise-firewalls-vs-open-source-a-comprehensive-comparison/>
[Accessed 23 March 2026].